
SISTEMAS WEB

CURSO 2025/2026

OAuth en DropBox

Crear una aplicación que acceda a la cuenta de Dropbox de un usuario y obtenga la lista de archivos contenidos en una carpeta determinada



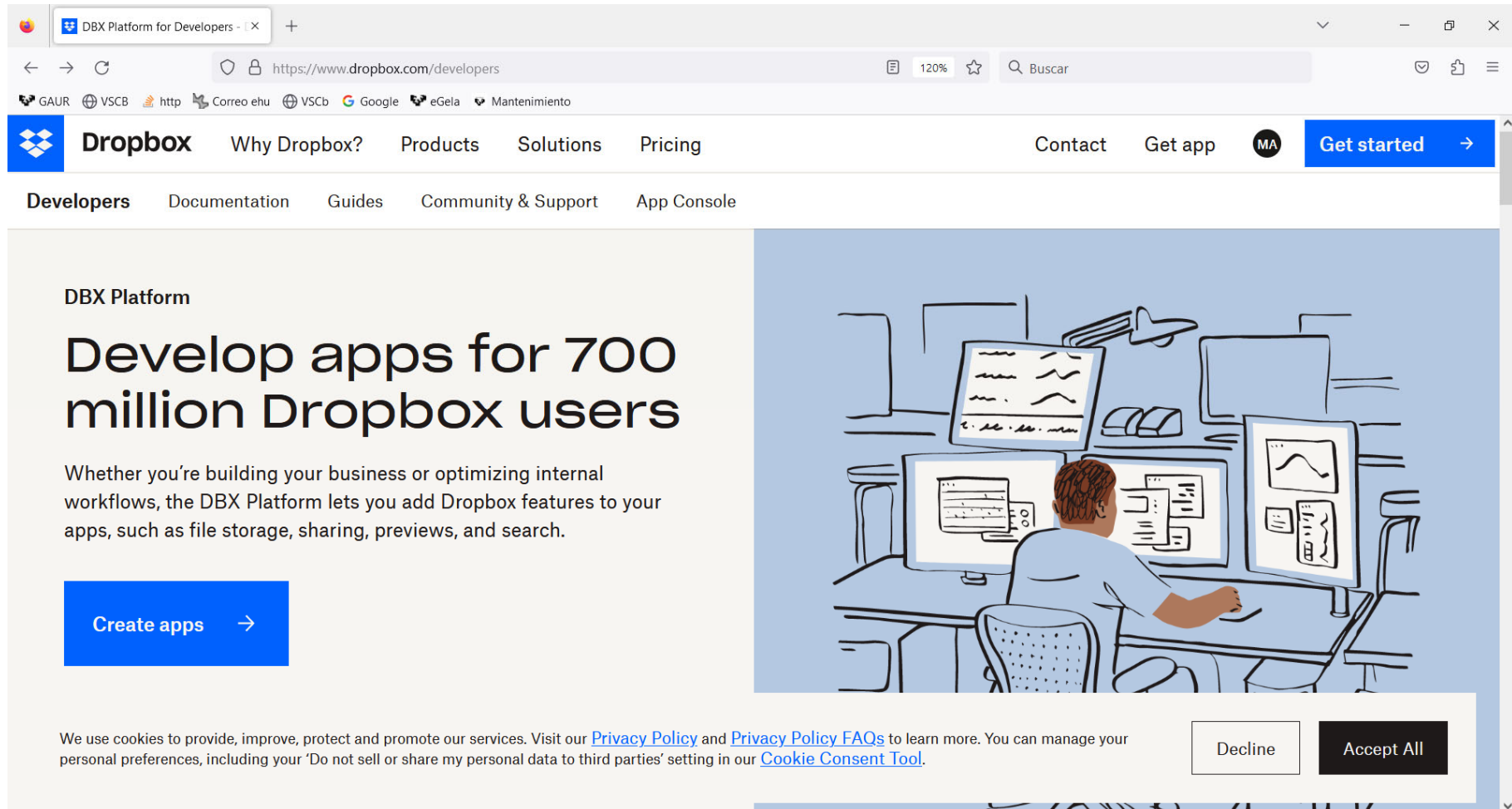
Sistemas Web by Oskar Casquero & María Luz Álvarez is licensed under a [Creative Commons Reconocimiento 4.0 Internacional License](https://creativecommons.org/licenses/by/4.0/).

OAUTH2 EN DROPBOX

1. **Registro de la Aplicación:** Antes de comenzar, necesitas registrar tu aplicación en el panel de desarrolladores de Dropbox. Durante este proceso, recibirás un *client_id* y un *client_secret* que identifican tu aplicación.
2. **Solicitud de Autorización:** Cuando un usuario quiere permitir que tu aplicación acceda a su cuenta de Dropbox, tu aplicación construye una **URL de autorización** que incluye tu *client_id*, y opcionalmente un *redirect_uri* y otros parámetros necesarios. Esta URL es la que se le presenta al usuario para que inicie sesión en Dropbox y autorice a tu aplicación.
3. **Autorización del Usuario:** Cuando el usuario hace clic en el enlace de autorización y se autentica en Dropbox, se le pedirá que apruebe los permisos que tu aplicación solicita. Una vez que el usuario aprueba estos permisos, Dropbox redirige al usuario de vuelta a tu aplicación, generalmente a la *redirect_uri* que especificaste, y pasa un **código de autorización (auth code)** en la URL.
4. **Intercambio del Código de Autorización por un Token de Acceso:** Tu aplicación debe tomar el código de autorización recibido y enviarlo junto con tu *client_id* y *client_secret* al punto final de token de Dropbox. Dropbox verificará la validez del código de autorización y, si es correcto, devolverá un **token** de acceso que tu aplicación puede usar para realizar solicitudes en nombre del usuario.
5. **Acceso a Recursos Protegidos:** Con el token de acceso obtenido, tu aplicación puede hacer solicitudes a la API de Dropbox en nombre del usuario. Cada solicitud debe incluir el token de acceso en la cabecera de autorización para demostrar que está autorizada.
6. **Renovación de Tokens:** Los tokens de acceso pueden caducar después de un cierto período de tiempo. Tu aplicación debe manejar la renovación de tokens de manera adecuada, solicitando un nuevo token de acceso utilizando el token de actualización proporcionado por Dropbox cuando el token de acceso caduque.

DESARROLLADORES DROPBOX

HTTPS://WWW.DROPBOX.COM/DEVELOPERS/



The screenshot shows the Dropbox Developers page in a web browser. The browser's address bar displays the URL <https://www.dropbox.com/developers>. The page features a navigation bar with the Dropbox logo and links for 'Why Dropbox?', 'Products', 'Solutions', 'Pricing', 'Contact', 'Get app', and a 'Get started' button. Below the navigation bar, there are links for 'Developers', 'Documentation', 'Guides', 'Community & Support', and 'App Console'. The main content area is titled 'DBX Platform' and features the headline 'Develop apps for 700 million Dropbox users'. A sub-headline states: 'Whether you're building your business or optimizing internal workflows, the DBX Platform lets you add Dropbox features to your apps, such as file storage, sharing, previews, and search.' A blue button labeled 'Create apps' with a right arrow is positioned below the text. To the right of the text is a large illustration of a person sitting at a desk with multiple computer monitors displaying various data and charts. At the bottom of the page, a cookie consent banner is visible, with 'Decline' and 'Accept All' buttons.

DBX Platform

Develop apps for 700 million Dropbox users

Whether you're building your business or optimizing internal workflows, the DBX Platform lets you add Dropbox features to your apps, such as file storage, sharing, previews, and search.

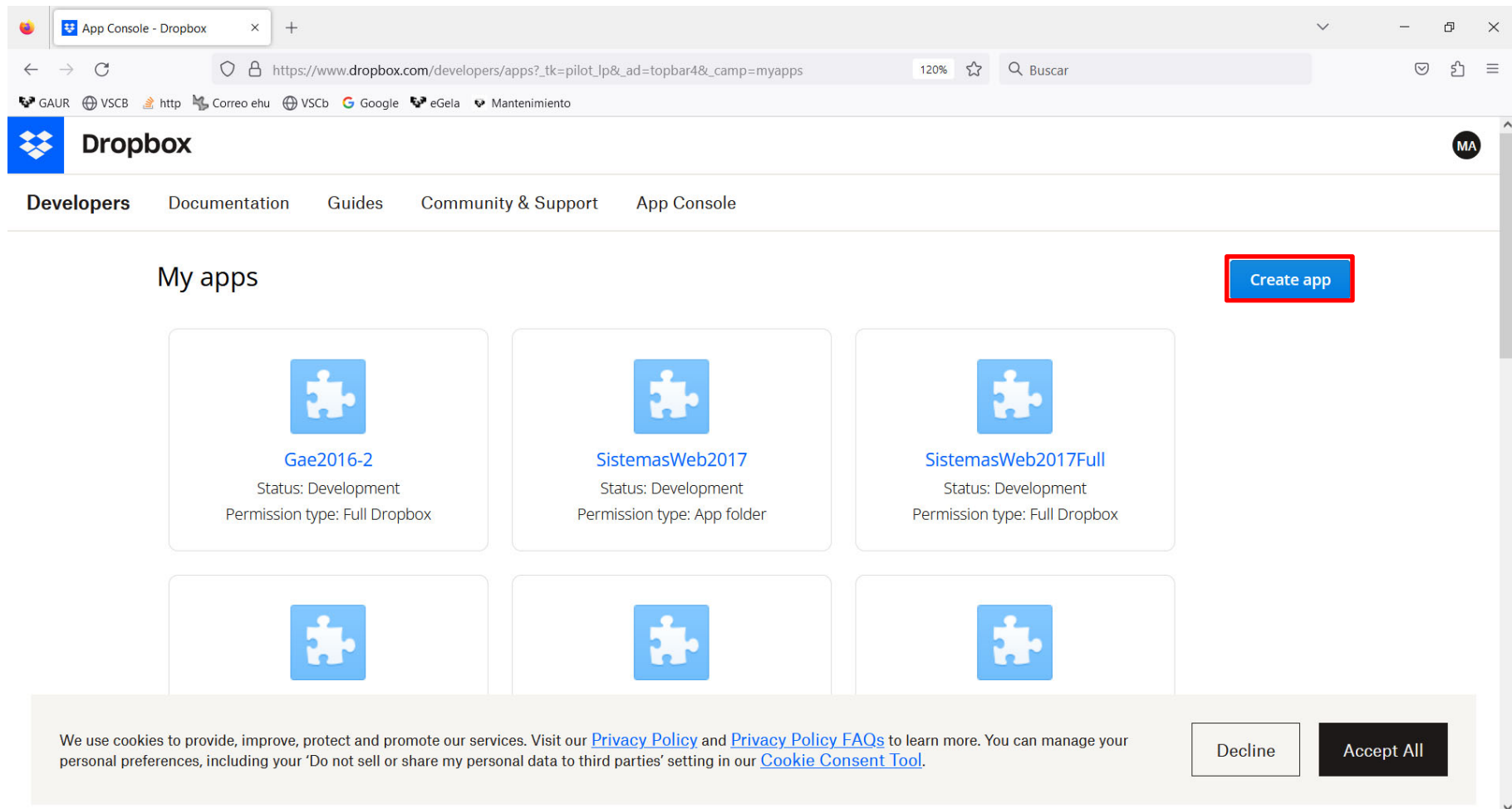
Create apps →

We use cookies to provide, improve, protect and promote our services. Visit our [Privacy Policy](#) and [Privacy Policy FAQs](#) to learn more. You can manage your personal preferences, including your 'Do not sell or share my personal data to third parties' setting in our [Cookie Consent Tool](#).

Decline Accept All

OAUTH DROPBOX APLICACIÓN

CREAR Y OBTENER CREDENCIALES DE LA APLICACIÓN



OAUTH DROPBOX APLICACIÓN

CREAR Y OBTENER CREDENCIALES DE LA APLICACIÓN

Developers - Dropbox

https://www.dropbox.com/developers/apps/create

70%

GAUR VSCB http Correo ehu VSCb Google eGela Mantenimiento

Dropbox

Developers Documentation Guides Community & Support App Console

Create a new app on the DBX Platform

1. Choose an API

Scoped access [New](#)

☒ Select the level of access your app needs to Dropbox data. [Learn more](#)

2. Choose the type of access you need

[Learn more about access types](#)

☒ App folder - Access to a single folder created specifically for your app.

☐ Full Dropbox - Access to all files and folders in a user's Dropbox.

3. Name your app

SW_2023

Create app

OAUTH DROPBOX APLICACIÓN

CREAR Y OBTENER CREDENCIALES DE LA APLICACIÓN

The screenshot shows the Dropbox Developers App Console for an application named 'SW_2023 - Dropbox'. The browser address bar shows the URL 'https://www.dropbox.com/developers/apps/info/'. The page has a navigation bar with links: Developers, Documentation, Guides, Community & Support, and App Console. A blue banner at the top says 'Add branding' with a subtext 'Give your users important information about your Dropbox app. Should comply with the Dropbox developer branding guide. Get started'. Below this, the app's status is 'Development' with a button 'Apply for production'. The 'Development users' section shows '0 / 500' with a button 'Unlink all users'. The 'Permission type' is 'Scoped App (App Folder)' with an info icon. The 'App folder name' is 'SW_2023' with a 'Change' button. A red rectangle highlights the 'App key' field with the value 'p8d4d801e597qev' and the 'App secret' field with a 'Show' link. Below this, the 'OAuth 2' section is visible, with a red arrow pointing to the 'Redirect URIs' field. The 'Redirect URIs' field contains 'http://localhost:8090' and a button 'Add'. Below that, there is a field for 'https:// (http allowed for localhost)' and an 'Add' button. At the bottom, the 'Allow public clients (Implicit Grant & PKCE)' section has a dropdown menu set to 'Allow'.

Archivo Editar Ver Historial Marcadores Herramientas Ayuda

SW_2023 - Dropbox

https://www.dropbox.com/developers/apps/info/ 70% dropbox develop

GAUR VSCB Google Mantenimiento eGela Correo: MARIA LUZ AL... Inicio - OneDrive CloudConsole

Dropbox

Developers Documentation Guides Community & Support App Console

Add branding
Give your users important information about your Dropbox app. Should comply with the Dropbox developer branding guide. [Get started](#)

Status [Development](#) [Apply for production](#)

Development users 0 / 500 [Unlink all users](#)

Permission type [Scoped App \(App Folder\)](#)

App folder name SW_2023 [Change](#)

App key p8d4d801e597qev
App secret [Show](#)

OAuth 2 **Redirect URIs**

[http://localhost:8090](#) [X](#)

[https:// \(http allowed for localhost\)](#) [Add](#)

Allow public clients (Implicit Grant & PKCE)

[Allow](#)

OAUTH DROPBOX APLICACIÓN

CREAR Y OBTENER CREDENCIALES DE LA APLICACIÓN

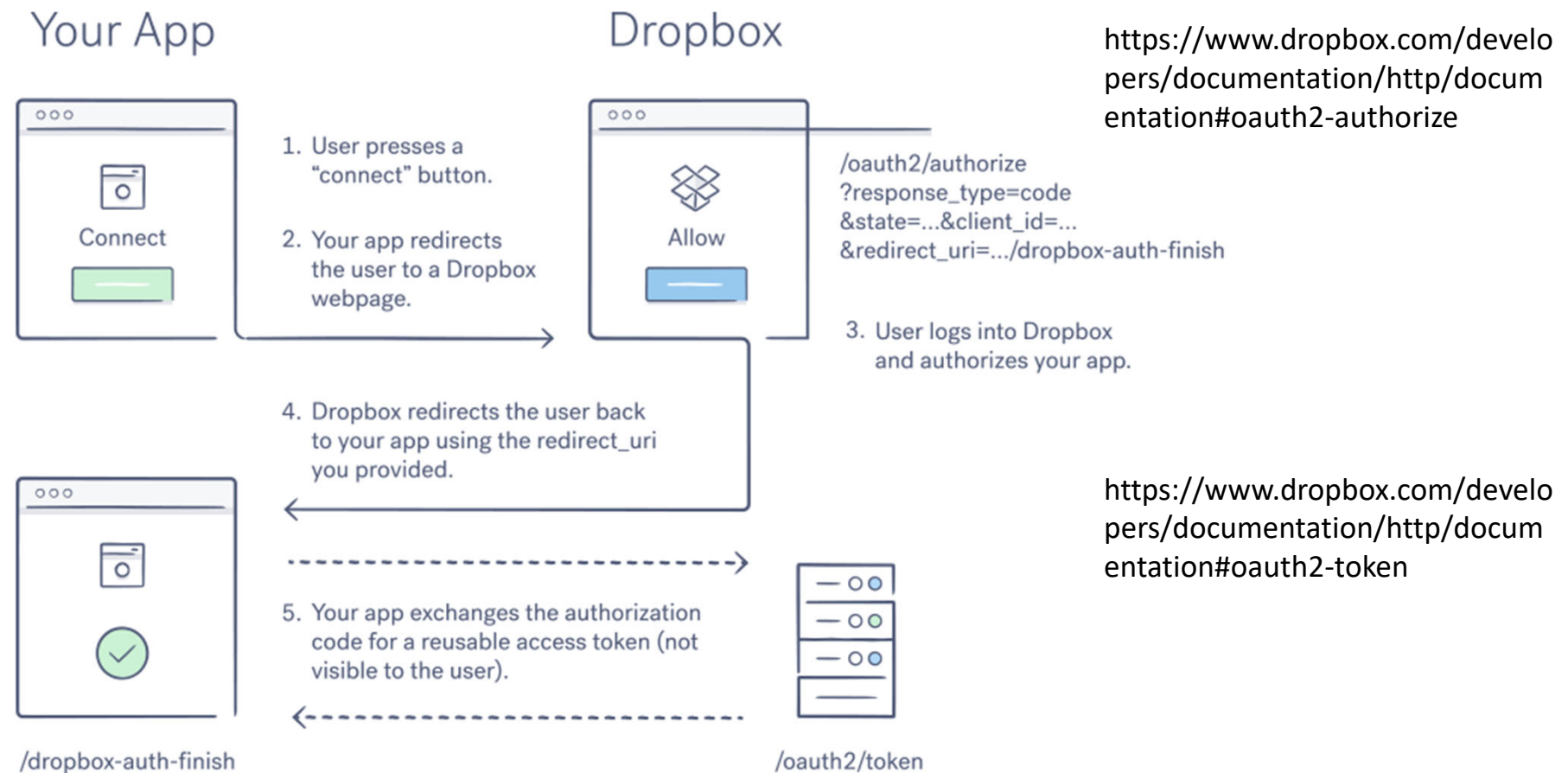
The screenshot shows the Dropbox Developers console for an application named 'SW_2023'. The 'Permissions' tab is selected and highlighted with a red box. The page displays various permission categories and their associated scopes:

- Individual Scopes**: Individual scopes include the ability to view and manage a user's files and folders. [View Documentation](#)
- Account Info**: Permissions that allow your app to view and manage Dropbox account info
 - ☐ `account_info.write`: View and edit basic information about your Dropbox account such as your profile photo
 - ☒ `account_info.read`: View basic information about your Dropbox account such as your username, email, and country
- Files and folders**: Permissions that allow your app to view and manage files and folders
 - ☐ `files.metadata.write`: View and edit information about your Dropbox files and folders
 - ☐ `files.metadata.read`: View information about your Dropbox files and folders
 - ☐ `files.content.write`: Edit content of your Dropbox files and folders
 - ☐ `files.content.read`: View content of your Dropbox files and folders
- Collaboration**: Permissions that allow your app to view and manage sharing and collaboration settings
 - ☐ `sharing.write`: View and manage your Dropbox sharing settings and collaborators
 - ☐ `sharing.read`: View your Dropbox sharing settings and collaborators
 - ☐ `file_requests.write`: View and manage your Dropbox file requests
 - ☐ `contacts.write`: View and manage your manually added Dropbox contacts

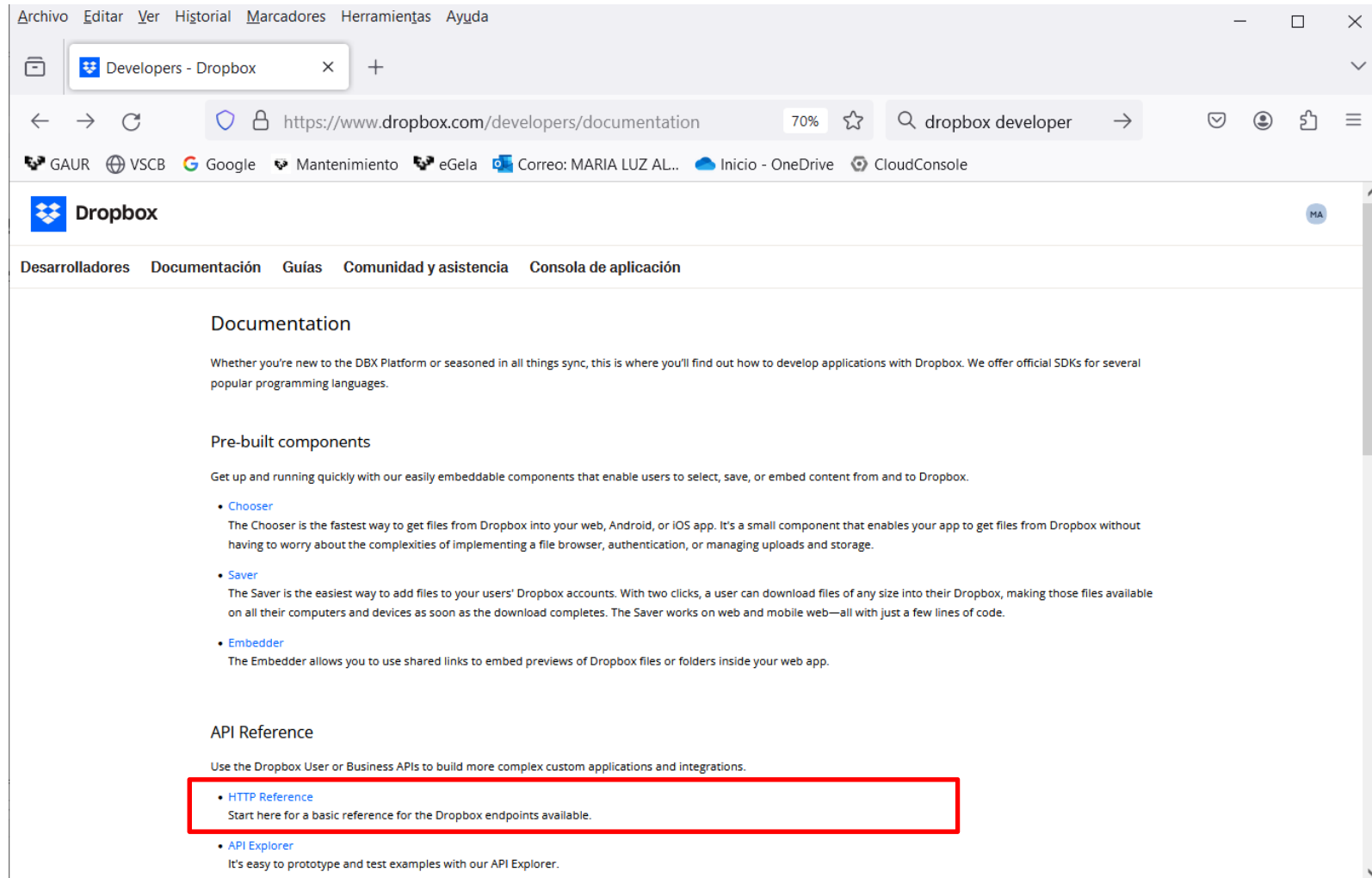
A black notification box at the bottom of the permissions list states: "Click Submit when you are done making changes. (Existing access tokens will not be affected)". The box contains "Undo" and "Submit" buttons.

OAUTH DROPBOX APLICACIÓN

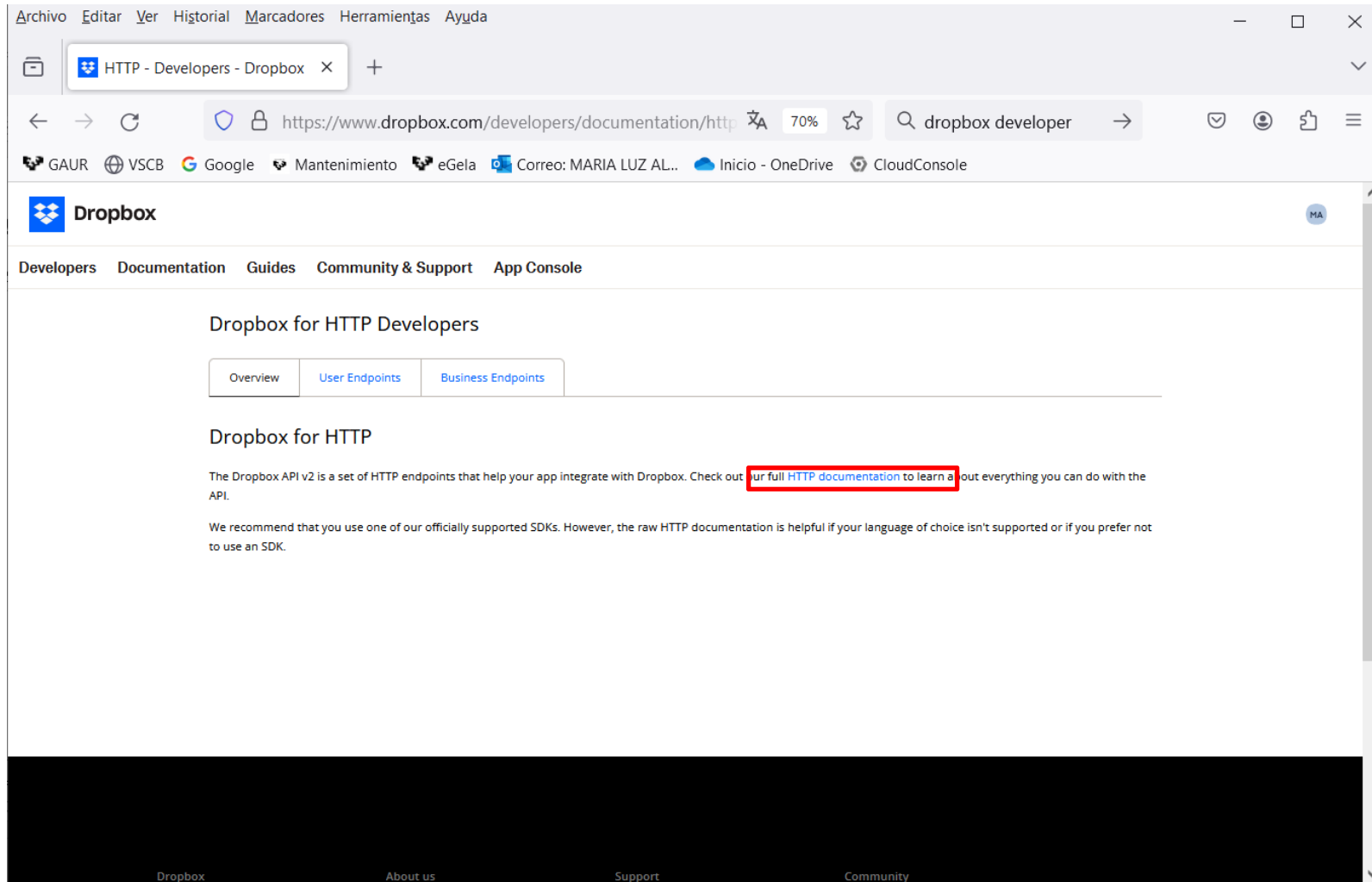
OAUTH CODE Y TOKEN



HTTPS://WWW.DROPBOX.COM/DEVELOPERS/DOCUMENTATION



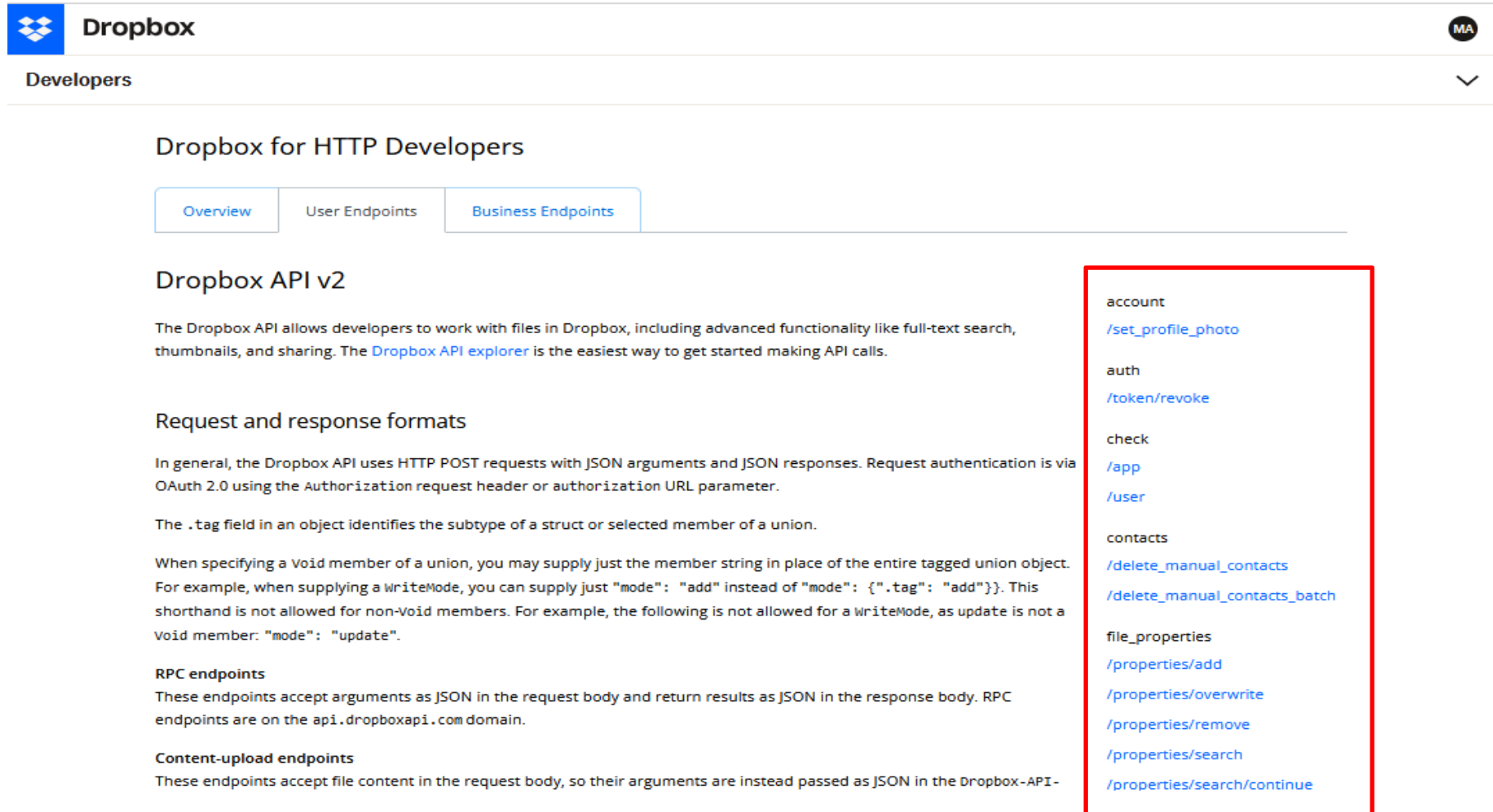
HTTPS://WWW.DROPBOX.COM/DEVELOPERS/DOCUMENTATION/HTTP/OVERVIEW



OAUTH DROPBOX APLICACIÓN

UTILIZAR LA API DE DROPBOX

<https://www.dropbox.com/developers/documentation/http/documentation>



Dropbox

Developers

Dropbox for HTTP Developers

[Overview](#) [User Endpoints](#) [Business Endpoints](#)

Dropbox API v2

The Dropbox API allows developers to work with files in Dropbox, including advanced functionality like full-text search, thumbnails, and sharing. The [Dropbox API explorer](#) is the easiest way to get started making API calls.

Request and response formats

In general, the Dropbox API uses HTTP POST requests with JSON arguments and JSON responses. Request authentication is via OAuth 2.0 using the Authorization request header or authorization URL parameter.

The `.tag` field in an object identifies the subtype of a struct or selected member of a union.

When specifying a Void member of a union, you may supply just the member string in place of the entire tagged union object. For example, when supplying a `WriteMode`, you can supply just `"mode": "add"` instead of `"mode": {".tag": "add"}`. This shorthand is not allowed for non-void members. For example, the following is not allowed for a `WriteMode`, as `update` is not a Void member: `"mode": "update"`.

RPC endpoints

These endpoints accept arguments as JSON in the request body and return results as JSON in the response body. RPC endpoints are on the `api.dropboxapi.com` domain.

Content-upload endpoints

These endpoints accept file content in the request body, so their arguments are instead passed as JSON in the Dropbox-API-

- `account`
 - [/set_profile_photo](#)
- `auth`
 - [/token/revoke](#)
- `check`
 - [/app](#)
 - [/user](#)
- `contacts`
 - [/delete_manual_contacts](#)
 - [/delete_manual_contacts_batch](#)
- `file_properties`
 - [/properties/add](#)
 - [/properties/overwrite](#)
 - [/properties/remove](#)
 - [/properties/search](#)
 - [/properties/search/continue](#)

OAUTH DROPBOX APLICACIÓN

UTILIZAR LA API DE DROPBOX



Developers Documentation Guides Community & Support App Console

`too_many_files` *Void* The operation involves more than 25 files.

See also [general errors](#).

`/list_folder`

VERSION

DESCRIPTION Starts returning the contents of a folder. If the result's `ListFolderResult.has_more` field is true, call `listFolderContinue()` with the returned `ListFolderResult.cursor` to retrieve more entries. If you're using `ListFolderArg.recursive` set to true to keep a local cache of the contents of a Dropbox account, iterate through each entry in order and process them as follows to keep your local state in sync: For each `FileMetadata`, store the new entry at the given path in your local state. If the required parent folders don't exist yet, create them. If there's already something else at the given path, replace it and remove all its children. For each `FolderMetadata`, store the new entry at the given path in your local state. If the required parent folders don't exist yet, create them. If there's already something else at the given path, replace it but leave the children as they are. Check the new entry's `FolderSharingInfo.read_only` and set all its children's read-only statuses to match. For each `DeletedMetadata`, if your local state has something at the given path, remove it and all its children. If there's nothing at the given path, ignore this entry. Note: `auth.RateLimitError` may be returned if multiple `listFolder()` or `listFolderContinue()` calls with same parameters are made simultaneously by same API app for same user. If your app implements retry logic, please hold off the retry until the previous request finishes.

URL STRUCTURE `https://api.dropboxapi.com/2/files/list_folder`

AUTHENTICATION [User Authentication](#), [App Authentication](#), [Dropbox-API-Select-Admin \(Whole Team\)](#)

ENDPOINT FORMAT [RPC](#)

REQUIRED SCOPE `files.metadata.read`

EXAMPLE Get access token for: [Sign in to pick app](#)

[files](#)

[/copy](#)

[/copy_batch](#)

[/copy_batch/check](#)

[/copy_reference/get](#)

[/copy_reference/save](#)

[/create_folder](#)

[/create_folder_batch](#)

[/create_folder_batch/check](#)

[/delete](#)

[/delete_batch](#)

[/delete_batch/check](#)

[/download](#)

[/download_zip](#)

[/export](#)

[/get_file_lock_batch](#)

[/get_metadata](#)

[/get_temporary_link](#)

[/get_temporary_upload_link](#)

[/get_thumbnail](#)

[/get_thumbnail_batch](#)

[/list_folder](#)

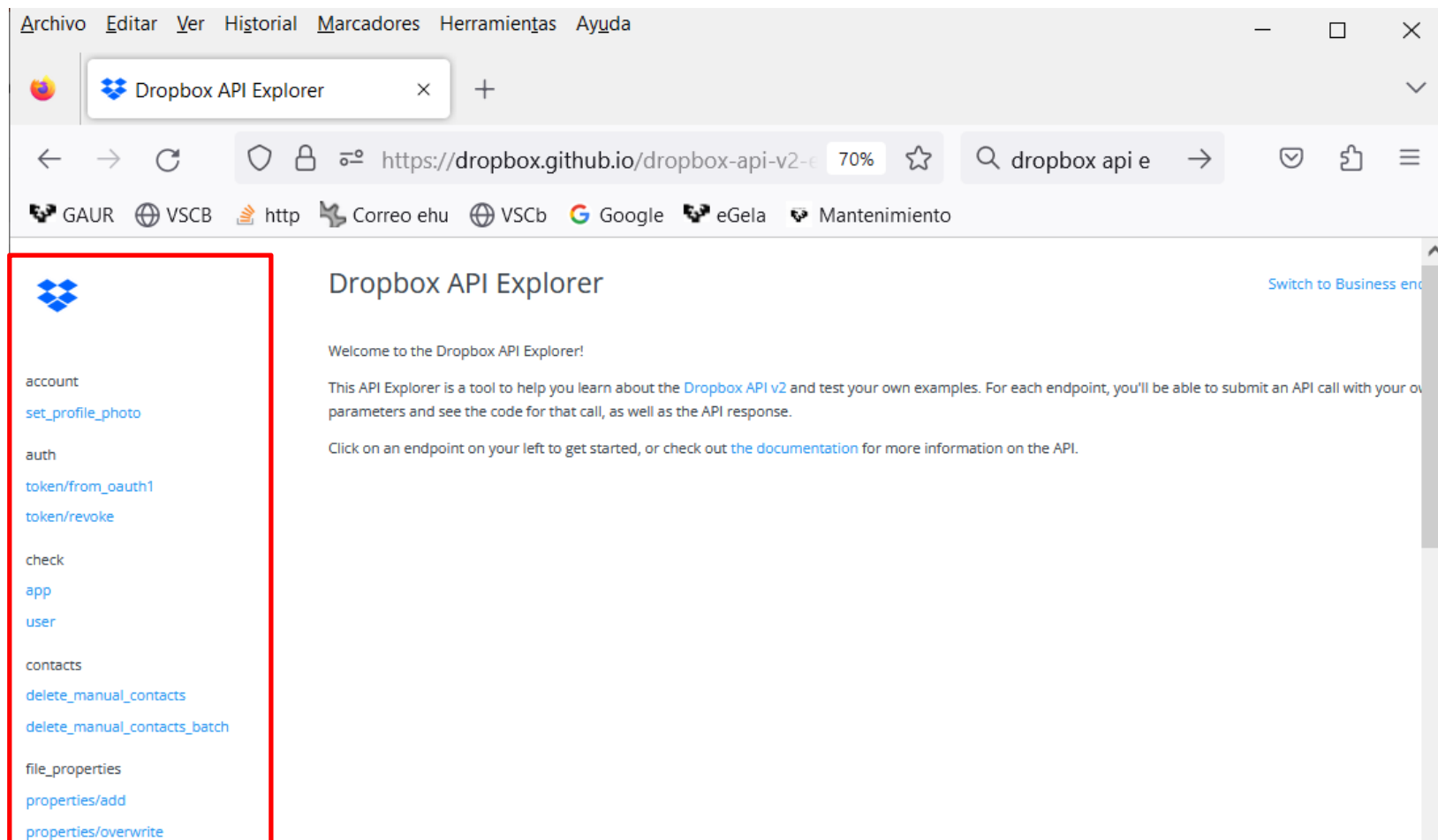
[/list_folder/continue](#)



OAUTH DROPBOX APLICACIÓN

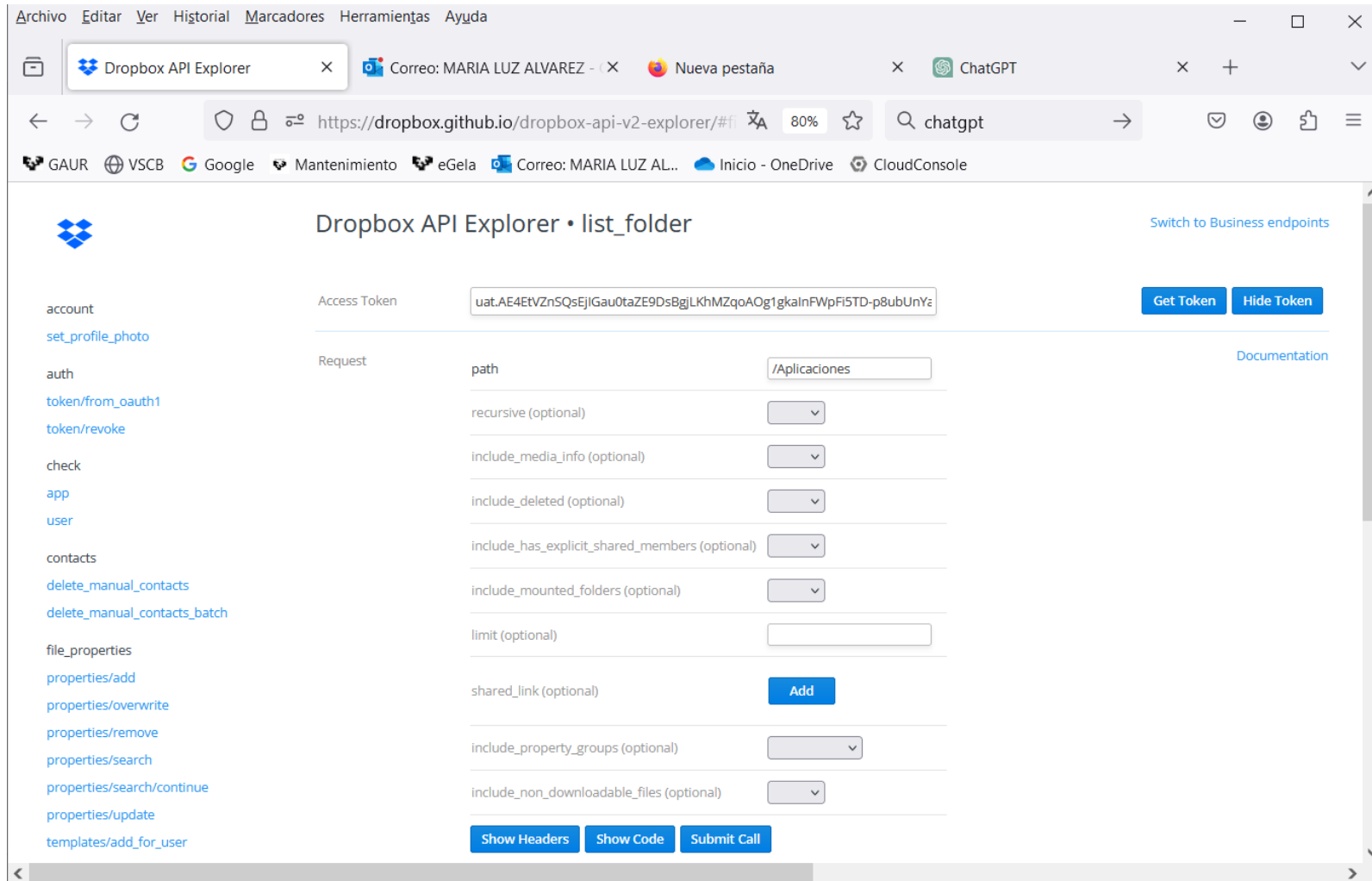
EXPLORADOR DE LA API

<https://dropbox.github.io/dropbox-api-v2-explorer>



OAUTH DROPBOX APLICACIÓN

EXPLORADOR DE LA API



OAUTH DROPBOX APLICACIÓN

EXPLORADOR DE LA API

Archivo Editar Ver Historial Marcadores Herramientas Ayuda

Dropbox API Explorer Correo: MARIA LUZ ALVAREZ - Nueva pestaña

https://dropbox.github.io/dropbox-api-v2-explorer/#fi 80%

GAUR VSCB Google Mantenimiento eGela Correo: MARIA LUZ AL... Inicio - OneDrive

properties/add
properties/overwrite
properties/remove
properties/search
properties/search/continue
properties/update
templates/add_for_user
templates/get_for_user
templates/list_for_user
templates/remove_for_user
templates/update_for_user

file_requests
count
create
delete
delete_all_closed
get
list_v2
list
list/continue
update
files

shared_link (optional) Add

include_property_groups (optional)

include_non_downloadable_files (optional)

Show Headers Hide Code Submit Call

Code View request as HTTP request

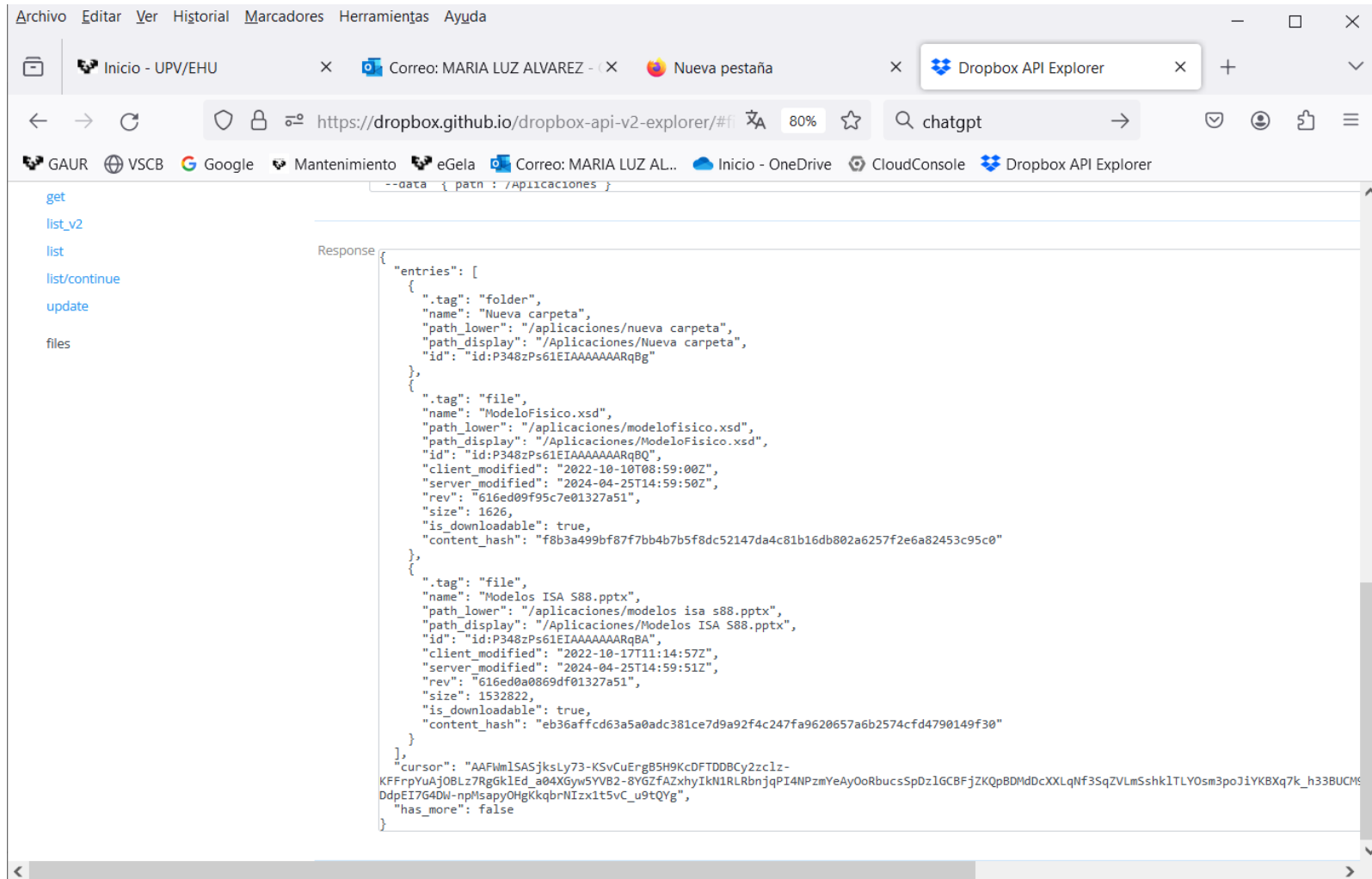
POST /2/files/list_folder
Host: https://api.dropboxapi.com
User-Agent: api-explorer-client
Authorization: Bearer uat.AE4EtVzn5QsEjIGau0taZE9DsBgjLKhMZqoA0g1gkaInFwpFi5TD-p8ubUnYasNXMEu01YrLCyJsZZjy-VkGwZ4jemSLAocysfgM4t-T8uZ80701soj0y1IVHo86VV6JGd8aBecXckrpUgRryMyitMeotRXDRRp35AKsL1bKp_ZjLuKYb7obH2mZicFCKK5zKs4u4FZz10Rx9Gg6INAaGkGqFctCd0mJXCw7unQepzxyhFcPtgTVPY1aurT_vRR4xVKwgp5kCce11F7DbwIbtqu0SiiYjetAjaVtd30GJGt7Q8tMV05qVgYSgeU6MV5C6RI1bfiJ7MD1wBGS1hpxB6tn3gHmChzoKsUVcvmbz_jBzBTNK_5EIEAzSqDY4uwKo0q6rlgaeH2RNBT_NlefzdPySfeIaIpWl0M1cF00ocFg7N6KeYfwtbtlqIP1MPAgIUWQKADnz84-thRq90iYxq04sVn1vMuJN_pAv4Fxiu0tH4EQm6_390GNspWEMdRUbwYpaApsKjgtuer062XjgoHAWfnja_wmdeK15Rrly8Jlti748nXqWQ_FCqyw0cWspuNlPAUAvyhs1GHMYa59W2RfdFeFBWJthk_uzz21nR7rFJ04oXq3fmh2Yd3kRr_2LvFs4TCW7zZSSiyF16tUXUzkzHwI1l7p9wpYm8VZdnPybq9Sfb_w4Hs2G0WqVyEl_rI7EeaYBQ-R8Rze3K04ipA0LVa3CZ9hLmE150S8620QVYXABqWjq9d0aDYyF40E3Tm73ctYFayIhqsYE2XwM1oNqjYgTRmYJvoBk9ZngFd3V0qdbMniSHcD6ZuwrPZ4aLq46C85k-VTnkOupndqKdK_D9hfwCChiynSvNrvWAZDwLCQsD1n8XXvhNQSMekohQkjNCRsELfpNgg4RUX9u1zQtNd81utfh16nOW1phWdzV2b2CtKmX5g40CkvytM5phT7ZDd_aw7mILqZ_vP5nNJR68VR1AhXk0eLA
Content-Type: application/json

{
 "path": "/Aplicaciones"
}

Response {
 "entries": [
 {
 "tag": "folder",
 "name": "Nueva carpeta",
 "path_lower": "/aplicaciones/nueva carpeta",
 "path_display": "/Aplicaciones/Nueva carpeta",
 "id": "id:P34RzP561FTAAAAAARnRpe"
 }
]
}

OAUTH DROPBOX APLICACIÓN

UTILIZAR LA API DE DROPBOX



The screenshot shows the Dropbox API Explorer interface. On the left, a sidebar lists API methods: get, list_v2, list, list/continue, update, and files. The main area displays a GET request to the endpoint `--data { path : /Aplicaciones }`. Below the request, the JSON response is shown, containing a list of application entries. The response includes details for a folder and two files, such as their names, paths, IDs, and modification timestamps.

```
Response {
  "entries": [
    {
      ".tag": "folder",
      "name": "Nueva carpeta",
      "path_lower": "/aplicaciones/nueva carpeta",
      "path_display": "/Aplicaciones/Nueva carpeta",
      "id": "id:P348zPs61EIAAAAAAARqBg"
    },
    {
      ".tag": "file",
      "name": "ModeloFisico.xsd",
      "path_lower": "/aplicaciones/modelofisico.xsd",
      "path_display": "/Aplicaciones/ModeloFisico.xsd",
      "id": "id:P348zPs61EIAAAAAAARqBQ",
      "client_modified": "2022-10-10T08:59:00Z",
      "server_modified": "2024-04-25T14:59:50Z",
      "rev": "616ed09f95c7e01327a51",
      "size": 1626,
      "is_downloadable": true,
      "content_hash": "f8b3a499bf87f7bb4b7b5f8dc52147da4c81b16db802a6257f2e6a82453c95c0"
    },
    {
      ".tag": "file",
      "name": "Modelos ISA S88.pptx",
      "path_lower": "/aplicaciones/modelos isa s88.pptx",
      "path_display": "/Aplicaciones/Modelos ISA S88.pptx",
      "id": "id:P348zPs61EIAAAAAAARqBA",
      "client_modified": "2022-10-17T11:14:57Z",
      "server_modified": "2024-04-25T14:59:51Z",
      "rev": "616ed0a0869df01327a51",
      "size": 1532822,
      "is_downloadable": true,
      "content_hash": "eb36affcd63a5a0adc381ce7d9a92f4c247fa9620657a6b2574cfd4790149f30"
    }
  ],
  "cursor": "AAFwmlSASjksLy73-KSvCuErgB5H9KcDFTDDBCy2zclz-KFFrPyUaj0BLz7RgGklEd_a04XGyw5YVb2-8YGZfAZxhyIkN1RLRbnjqPI4NPzmYeAyOoRbucssSpDz1GCBFjZKQpBDMdDcXXLqNf3SqZVLmSshk1TLYOsm3poJ1YKBXq7k_h33BUCM9DdpE17G4DW-npMsapyOHgKkqbrNIzx1t5vC_u9tQYg",
  "has_more": false
}
```


OAUTH DROPBOX APLICACIÓN

AUTENTICACIÓN Y AUTORIZACIÓN DEL USUARIO

```
#-*- coding: utf-8 -*-
```

```
import urllib.parse
import requests
import webbrowser
from socket import AF_INET, socket, SOCK_STREAM
import json
```

```
app_key = 'xxxxxxxxxxxxxxxx'
app_secret = 'xxxxxxxxxxxxxxxx'
redirect_uri = "http:// localhost :8090"
```

```
#####1#####
# CODE: Abrir en el navegador la URI https://www.dropbox.com/oauth2/authorize #
#####
```

```
servidor = 'www.dropbox.com'
params = {'response_type': 'code',
          'client_id': app_key,
          'redirect_uri': redirect_uri }
```

```
params_encoded = urllib.parse.urlencode(params)
recurso = '/oauth2/authorize?' + params_encoded
uri = 'https://' + servidor + recurso
webbrowser.open_new(uri)
```

OAUTH DROPBOX APLICACIÓN

AUTENTICACIÓN Y AUTORIZACIÓN DEL USUARIO

```
# Crear servidor local que escucha por el puerto 8090
server_socket = socket(AF_INET, SOCK_STREAM)
server_socket.bind(('localhost', 8090))
server_socket.listen(1)
print("\tLocal server listening on port 8090")

# Recibir la solicitud 302 del navegador
client_connection, client_address = server_socket.accept()
peticion = client_connection.recv(1024)
print("\tRequest from the browser received at local server:")

# Buscar en la petición el "auth_code"
primera_linea = peticion.decode('UTF8').split('\n')[0]
print(primera_linea)
aux_auth_code = primera_linea.split(' ')[1]
auth_code = aux_auth_code[7:].split('&')[0]
print ("\tauth_code:" + auth_code)

# Devolver una respuesta al usuario
http_response = "HTTP/1.1 200 OK\r\n\r\n" \
    "<html>" \
    "<head><title>Prueba</title></head>" \
    "<body>The authentication flow has completed. Close this window.</body>" \
    "</html>"

client_connection.sendall(http_response.encode(encoding="utf-8"))
client_connection.close()
server_socket.close()
```

OAUTH DROPBOX APLICACIÓN

PROCESO DE OBTENCIÓN DEL TOKEN

```
#####  
# ACCESS_TOKEN: Obtener el TOKEN https://www.api.dropboxapi.com/1/oauth2/token #  
#####
```

```
params = {'code': auth_code,  
          'grant_type': 'authorization_code',  
          'client_id': app_key,  
          'client_secret': app_secret,  
          'redirect_uri': redirect_uri}  
  
cabeceras={'User-Agent': 'Python Client',  
           'Content-Type': 'application/x-www-form-urlencoded'}  
uri='https://api.dropboxapi.com/oauth2/token'  
respuesta = requests.post( uri, headers=cabeceras,data=params)  
  
print (respuesta.status_code)  
json_respuesta = json.loads(respuesta.content)  
access_token = json_respuesta['access_token']  
print ("Access-Token:" + access_token)
```

OAUTH DROPBOX APLICACIÓN

UTILIZAR LA API DE DROPBOX

```
print("/list_folder")

uri = 'https://api.dropboxapi.com/2/files/list_folder'
path= "/Aplicaciones"
datos = {'path': path}
datos_encoded = json.dumps(datos)

print("Datuaak: " + datos_encoded)
cabeceras = {'Host': 'api.dropboxapi.com',
             'Authorization': 'Bearer ' + access_token,
             'Content-Type': 'application/json'}
respuesta = requests.post(uri, headers=cabeceras, data=datos_encoded, allow_redirects=False)

status = respuesta.status_code
print ("\tStatus: " + str(status))
contenido = respuesta.text
print("\tContenido:")
contenido_json = json.loads(contenido)
print("Ficheros en " + path)
for entrie in contenido_json["entries"]:
    print(entrie['name'])
```

```
POST /2/files/list_folder HTTP/1.1
Host: https://api.dropboxapi.com
Authorization: Bearer access_token
Content-Type: application/json
Content-Lenght=TAMAÑO DEL CUERPO

{ "path": "/Aplicaciones" }
```